

Day 5 — Assignment

Numpy Fundamentals — arrays, broadcasting, aggregation, and a pivot table

Work through these tasks using only what we covered today. Target time: 30–45 minutes. Sections 1–3 use plain Numpy arrays you'll build yourself; Section 4 brings pandas back in, using winequality-white.csv.

Section 1 — Array Creation & Casting

Task 1: Create an array from the list [2, 4, 6, 8, 10]. Then create a second array from the same values as a tuple instead, explicitly forcing dtype to np.int32. Check both dtypes.

Task 2: Create an array mixing strings and numbers: ["3", 7, 9, "12", 5]. Check its dtype and explain what happened. Then force dtype=np.int32 on the same values — it should work. Finally, swap "12" for "girl" and force dtype=np.int32 again — what happens this time?

Task 3: Create: a 3x5 array of zeros, a 2x6 array of ones, a 5x5 identity matrix, and an array running from 0 to 20 in steps of 2.5.

Section 2 — Math & Broadcasting

Task 4: Create two 1D arrays of 4 numbers each. Add them together, then multiply them together, element-wise.

Task 5: Create a 2x2 array and multiply every element by 10 using broadcasting.

Task 6: Create a different 2x2 array. Subtract [1, 2] from it (column-wise broadcasting), then subtract [[1], [2]] from it (row-wise broadcasting). Explain the difference in the results.

Section 3 — Indexing & Aggregation

Task 7: Build a 4x5 array using arange() and reshape(), containing the numbers 0 through 19. Grab row 1 on its own, then column 3 on its own, then the sub-block spanning rows 1–2 and columns 2–3.

Task 8: On that same array, compute: the sum along axis=0, the mean along axis=1, the max along axis=0, and the min along axis=1.

Task 9: Find the index of the minimum value in each column (axis=0), and the index of the single largest value in the entire array.

Task 10: Compute the cumulative sum along `axis=1`, and then flatten the array back into one dimension.

Section 4 — `np.where()` and a Pivot Table

Task 11: Using the same 4x5 array from Section 3, find the positions of every value greater than 15, and separately, the position of the value 7.

Task 12: Read `winequality-white.csv`. Use `pd.cut()` to split the `alcohol` column into 5 equal-width bins, name the new column `alcohol_bin`, and attach it to the DataFrame. Then build a pivot table showing average residual sugar, with `quality` as the rows and `alcohol_bin` as the columns.